

Limitations:

- More complex to set up and manage than HPA.
- May not scale quickly enough for sudden spikes in resource demand.

Implementing autoscaling in EKS

Deploying the Cluster Autoscaler

- Use the official Helm chart or Kubernetes manifest to deploy the Cluster Autoscaler in your EKS cluster.
- Configure the Cluster Autoscaler to target specific node groups or namespaces.

Configuring autoscaling groups

- Create autoscaling groups for your EKS worker nodes.
- Configure the ASG settings like instance type, scaling policies, and min/max limits.
- Associate the ASGs with the Cluster Autoscaler.

Monitoring and optimisation

- Use tools like Amazon CloudWatch

or Prometheus to monitor cluster resource usage and scaling activity.

- Fine-tune the Cluster Autoscaler configuration based on resource needs and workload patterns.
- Consider implementing HPA for specific workloads requiring finer-grained scaling control.

The following are the components shown in Figure 1.

Pods: Represent the application workloads running in the cluster.

HPA: Monitors resource usage and scales pods within a deployment.

Cluster Autoscaler: Monitors unschedulable pods and triggers node provisioning/termination.

Autoscaling groups: Manage the life cycle of EC2 instances based on Cluster Autoscaler instructions.

CloudWatch: Provides monitoring and insights into resource utilisation and scaling activity.

Amazon EKS Anywhere for a hybrid cloud platform

Cloud service providers like Microsoft Azure, AWS and Google Cloud have been introducing new

features for multi-cloud and hybrid cloud architecture.

Google Anthos and Azure Arc support hybrid cloud architectures, allowing cloud service deployment on private clouds like VMWare or OpenShift. Amazon also offers a similar capability through Outposts, which was originally designed for AWS infrastructure but extended for private cloud platforms thus enabling AWS distribution like EC2 clusters on private cloud environments.

Now, AWS has tried to overcome the demerits of Outposts (Table 1) with a more advanced service that uses AWS EKS Anywhere. The latter extends Elastic Kubernetes Service (EKS) distribution for use in private cloud infrastructure on a custom data centre environment. So far, we have been able to deploy EKS services in a data centre using EKS Distro (EKS-D), which can run control planes on custom infrastructure, and also help run data planes on existing custom virtual machines or bare metal servers.

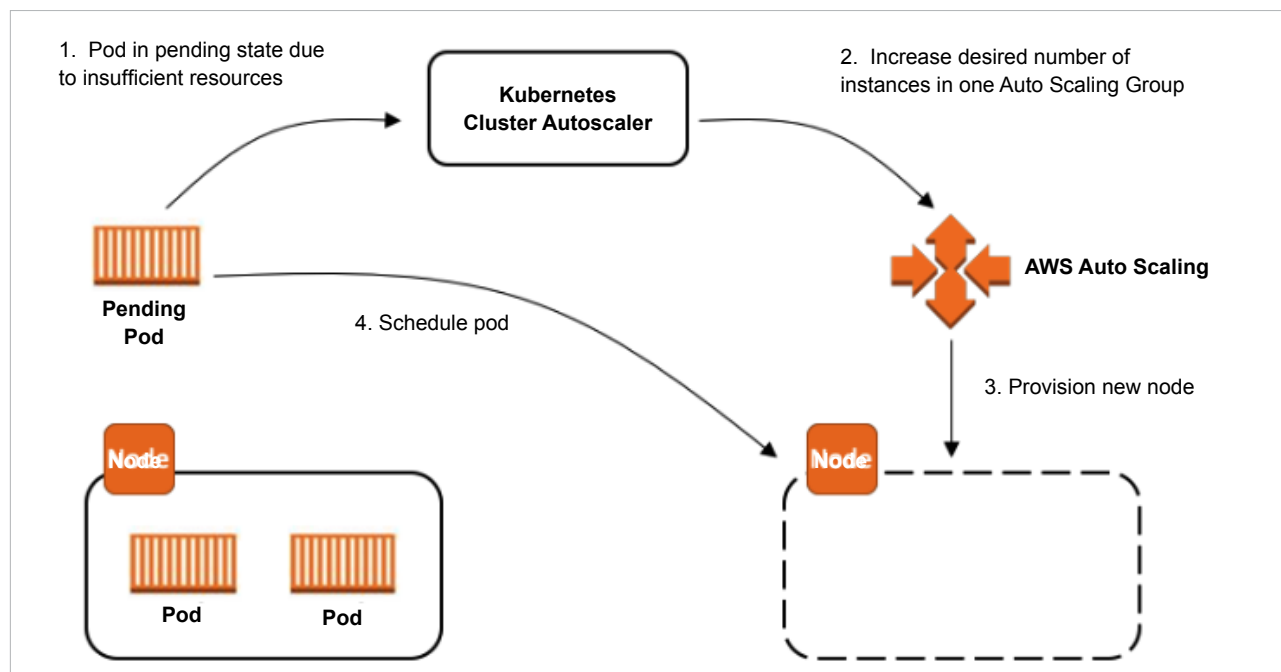


Figure 1: Cluster Autoscaler in EKS